



Geographic Information Support Team





Data and Information in Emergencies

- From early phases of an emergency information is critical
- No single organization or entity has all of this information
- Time is of the essence
- Integration of cross sectoral, institutional and geographical data increases the quality of decision making



Key Issues in Humanitarian Information

- User Requirements
- Quality of Data & Information
- Partnerships
- Preparedness
- Technology



Ongoing Activities of H.I.M.S.

- Geographic Information Support Team (GIST)
- SHARE
- Institutional Collaboration
- Field Support



Geographic Information Support Team

- What is it?
 - It is a group of technical experts
- Who are they?
 - UNHCR
 - UNICEF
 - FAO
 - WFP
 - OCHA
 - WB
 - WHO
 - OFDA
 - ECHO/JRC
 - DPKO



Geographic Information Support Team

- What do they do?
 - Provide a forum for geographic and geo-referenced information and data exchange amongst humanitarian response agencies and donors.
 - Promote the use of geo-referencing to enhance data and information co-ordination to improve humanitarian response.
 - Identify specialized resources and geographic information available/required to enhance support for potential and ongoing emergency operations.



The SHARE Approach

- A systematic approach to organizing critical information so that it can be pooled, analyzed, compared, contrasted, validated, reconciled, and mapped.
- SHARE provides a simple format for ensuring that data collected
- can be used by someone other than the originator.
- Based on 1) Geo-referenced information. 2) a time stamp 3) meta-data.



Institutional Collaboration

- Rapid HIC Deployment – (DFID)
- DPKO – Joint Geo–Cells
- Columbia Univ. – Hot Spots
- UNJLC – Field and HQ
- Agency Site - Rooms Joint Activation
- MAC Collaboration



Guidelines

- Use of remote sensing in humanitarian operations (developed by UNHCR / GIMU on behalf of the GIST)
- Humanitarian Information Centres (completed)
- The use of common standards - position and sectoral codes, (under completion)
- The development of Unique disaster Identifiers for disaster events world wide – (under completion)
- Use of GIS for disaster response, a practical guide



Field Support

- Existing:
 - The Kosovo HCIC
 - Information Coordination Centre in Eritrea (ICC)
 - Sierra Leone Humanitarian Information Center (SLIS)
 - The DEPHA in Nairobi
 - Humanitarian Information Center for Iraq
 - The AIMS for Afghanistan
 - The center in OPT
- Under Development:
 - The Southern Africa Hum. Info Center (SAHIM)
 - The regional center for South America – Panama
 - The planned center in Bangkok – WFP
 - The planned center in DPRK



Activation of Support

- When: At the onset of a natural or humanitarian crisis
- Who: Institutions represented in the GIST
- How : Conference call



Resources

- Financial
- Human
- Technological
- Data
- Know-How



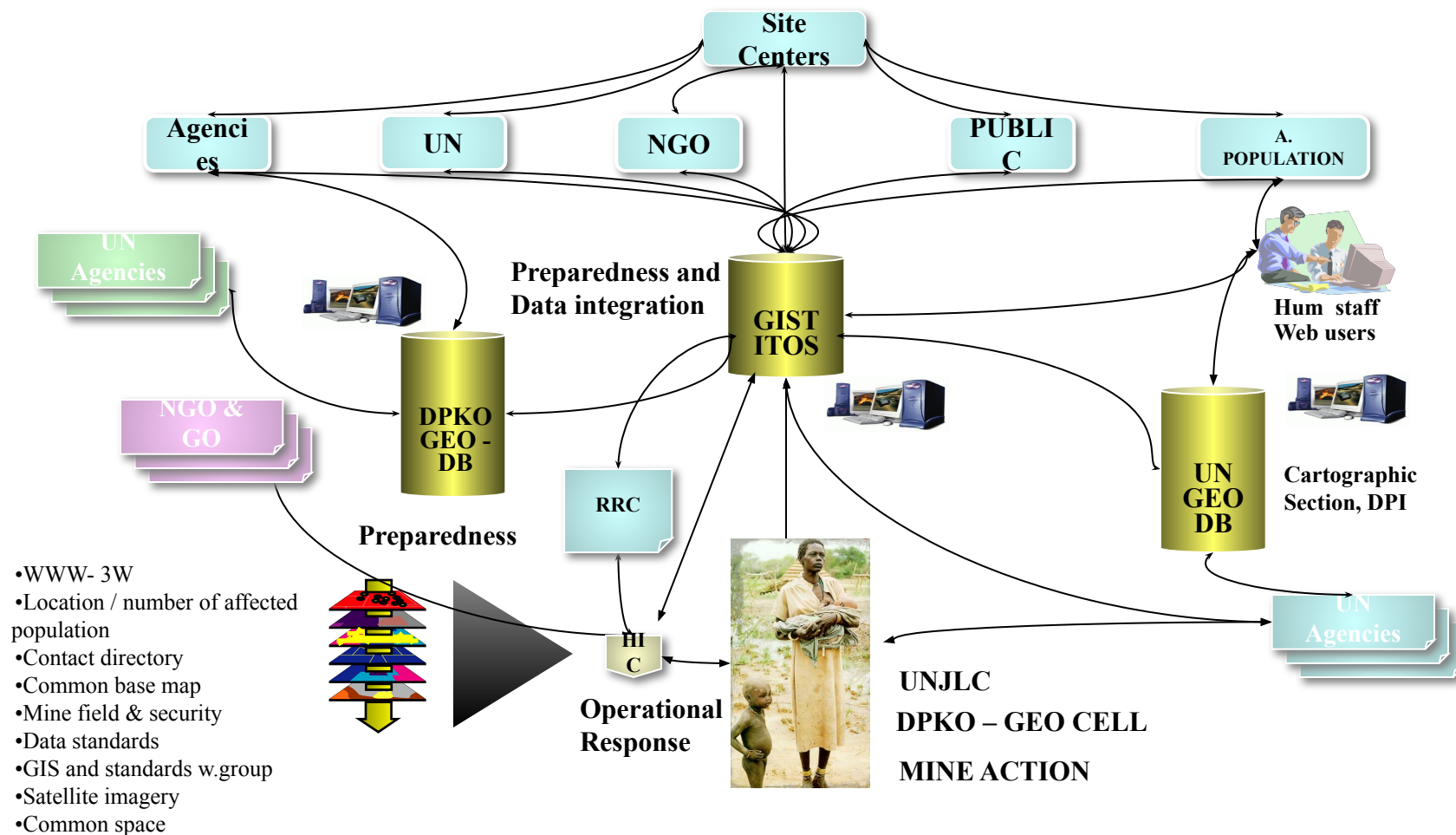
Data Integration and Dissemination

- GIST/ITOS - data repository and data integration from regional resource repositories and preparedness.
- The JRC – EU/ ECHO: Dig. Map Archive and EU-GIST site.
- The UN – Geo- data repository (UNGIWG)
- DPKO - Geo – data repository
- UNJLC – inter agency logistical web site
- The web - specific filed Sites – several organizations: DEPHA, Somalia info unit, AIMS, Kosovo.



Operational Concept

Preparedness and linkage to strategic coordination and operational areas at HQ



Humanitarian Crisis or Natural Disaster



Information Management

- Data preparedness is essential
- Common data standards
- Geographic data as an integrator
- Metadata must be acquired up-front



GIST Data Repository



Data Repository of the Geographic Information Support Team

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1 Draw search area ☐ :



2 Choose Data Type:
<All Data Types>
Choose Data Theme:
<All Data Themes>
Optional Keyword (e.g. river):
liberia
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Content Found by Search

Downloadable Data Sets

Content Title: [Central Monrovia & Mamba Point](#)
Data Originator: DigitalGlobe
Data Distributor: Geographic Information Support Team
Resolution: X Axis - 0.000007, Y Axis - 0.000007
Data Type: Raster

[View Details](#)

Content Title: [Liberia Administration 2 Boundaries](#)
Data Originator: Unknown
Data Distributor: Geographic Information Support Team
Data Type: Vector

[View Details](#)

Content Title: [Liberia Administration Boundaries Level 3](#)

Mined Areas





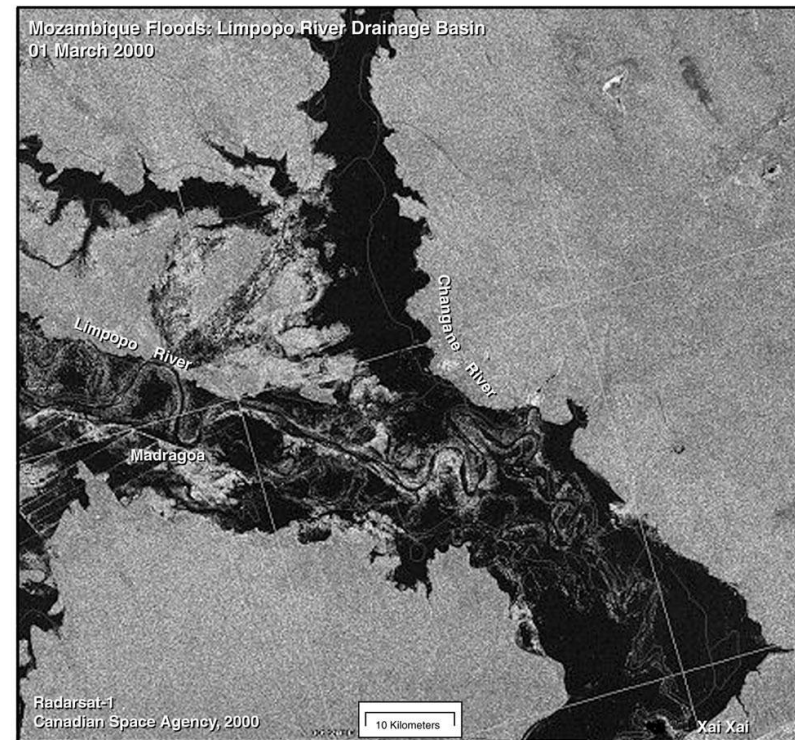
Operational Data (Rapid Needs Assessment)

1. LOCATION DETAILS:			
LOCATION:	ADMIN LEVEL A:	ADMIN LEVEL B:	P_CODE:
2. ASSESSMENT TRIP DETAILS:			
Organisation conducting assessment:	Person completing form:	Date (dd/mm/yy):	Location/Coordinates:
			Coordinates type:
3. LOCAL AUTHORITIES/CONTACTS:			
Name	Title	Contact number and office	
4. POPULATION INFORMATION:			
Previous population at location:	Current population at location:	Number of IDPs among current population:	
5. LOCATION TYPE:			
Location type (one only):		Is this location serving as distribution point:	
☐ Established settlement		☐ Yes	
☐ IDP Camp/Center		☐ No	
6. SECURITY SITUATION:			
Expected mine in the area:	General security situation:	Recent security related incidents aimed at humanitarian agencies/workers?	
☐ Yes			
☐ No			
7. TRANSPORTATION:			
Road type:	Road accessible by:	Obstacles on road:	Location also accessible by:
☐ Paved-Smooth	☐ Car		☐ Water
☐ Paved-Rough	☐ 4-wheel drive		☐ Air
☐ Dirt-Smooth	☐ Light truck		☐ Rail
☐ Dirt-Rough	☐ Heavy truck		
☐ Other:	☐ Other:		
8. ELECTRICITY:			
☐ Fully functional		☐ Intermittent ☐ Not functional	
How many hours per day:			

	A	B	C	D	K	L	M	N	O	P
	Provincia	Distrito	Location	C_code	Type	Population	Source	Date	House occupancy	Housing/shelter condition
2	Gaza	Chibuto	Calanga		Established village	10000	Iris	19-Mar-00		
3	Gaza	Chibuto	Samora Machel	090404	Temporary shelter	2922	CMOC		NA (tents)	NA (tents and
4	Gaza	Chibuto	Samora Machel	090404	Temporary shelter	3000	Malteser	15-Mar-00	NA (abandoned farm)	Broken windows,
5	Gaza	Chibuto	Samora Machel	090404	Temporary shelter			13-Mar-00	NA (tents)	NA (tents and
6	Gaza	Chókwe	25 de Setembro	090702	Established village	2674	CMOC	17-Feb-00	Most houses occupied	Houses either
7	Gaza	Chókwe	25 de Setembro	090702	Established village	2500	CONCERN	11-Mar-00	the houses occupied	damage. Dorms,
8	Gaza	Chókwe	Bombofo	090703	Established	945	CONCERN	11-Mar-00	Most houses empty	Over 30% roof
9	Gaza	Chókwe	Chate	090705	Established village	3600	CMOC	17-Mar-00	Most houses occupied	Houses with
10	Gaza	Chókwe	Chinacanine	090708	Temporary shelter	7390	CONCERN	11-Mar-00	NA (tents)	shelter
11	Gaza	Chókwe	Zuza	090724	village/temporary	4688		13-Mar-00		
12	Gaza	Chókwe	Conhane	090729	Established village	1000	UNDAC	26-Mar-00	occupied, only about	by floods, need
13	Gaza	Chókwe	Djodjo	090709	Established village	2500	CMOC	16-Mar-00	Most houses occupied	damage, doors
14	Gaza	Chókwe	Hokwe	090711	village/temporary	30650	ADPP	16-Mar-00	makeshift shelters	
15	Gaza	Chókwe	Hokwe	090711	Established village		UNICEF		Most houses occupied	Most houses are n
16	Gaza	Chókwe	Kumba		Established village	776	ACT/LWF	15-Mar-00	Most houses occupied	damaged
17	Gaza	Chókwe	Machinha		villages	2800	Purse	16-Mar-00		homes are totally
18	Gaza	Chókwe	Majajamela	090723	Established village	998	CMOC	17-Feb-00	Most houses occupied	completely
19	Gaza	Chókwe	Manjangue	090713	village/temporary	4185	CONCERN	11-Mar-00	the houses occupied	
20	Gaza	Chókwe	Marks River Camp	090728	Temporary shelter	500	Christ	13-Mar-00	NA (tents)	shelters)
21	Gaza	Chókwe	Marks River Camp	090728	Temporary shelter	456	Christ	13-Mar-00		
22	Gaza	Massingir	Tchaque/Obrigado Tanzania	090902	Established village	1800		13-Mar-00	Most houses occupied	Not damaged
23	Gaza	Guijá	7 de Abril	090301	Established village	2961	UNDAC	26-Mar-00	Most houses occupied	destruction in low
24	Gaza	Guijá	7 de Abril	090301	village/temporary	13498	ACT/LWF	15-Mar-00	Most houses occupied	homes destroyed
25	Gaza	Guijá	7 de Abril	090301	Established village	1200	UNICEF	02-Apr-00	living in tents	destroyed in 5
26	Gaza	Guijá	Canicado/Guijá Cidade	090302	village/temporary	7000		13-Mar-00		
27	Gaza	Guijá	Canicado/Guijá Cidade	090302	Established village	7500	Purse	20-Mar-00		homes not
28	Gaza	Guijá	Chivonguene	090304	Established village		UNICEF		occupied, the other	destroyed, other
29	Gaza	Guijá	Chibabele/Javanhane	090303	Established village	5057	ACT/LWF	15-Mar-00	occupied, not	be pretty much
30	Gaza	Guijá	Tomanine	090305	Established village	10000	ACT/LWF	15-Mar-00	Most houses occupied	destroyed
31	Gaza	Guijá	Tomanine	090305	Established village	4800	UNICEF		Most houses occupied	houses seriously
32	Gaza	Massangir	Macaringue	090904	Established village	1841	UNDAC	26-Mar-00	Most houses occupied	destroyed, other
33	Gaza	Massangir	Tchanue/Obrigado Tanzania	090902	Established village	1028	Purse	16-Mar-00		homes destroyed

Database

Form



Geographic Data

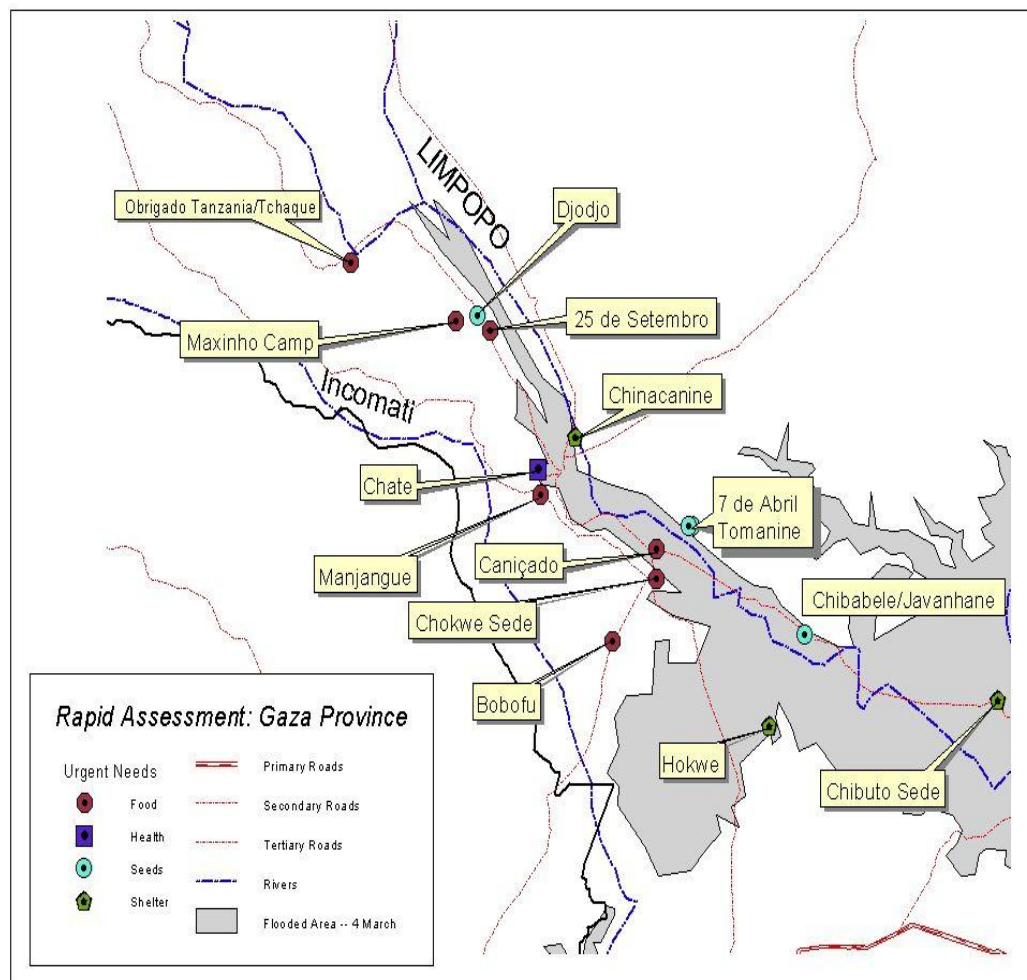
Satellite Imagery

Needs Locality

Locality	Sector	Date	Indicator
<u>Beilinson</u>	No Food Items Nutrition	28/06/2000 28/06/2000	Monitoring required Lack of Milk
<u>Geneva</u>	Agricultural input Protection	04/07/2000 24/07/2000	Fertilizant Distribution Abuse of Women
<u>Grand Pre</u>	Water & Sanitation	28/06/2000	Lack of Wells
<u>Lausanne</u>	Nutrition Nutrition Health	28/06/2000 28/06/2000 29/06/2000	Lack of Milk Monitoring Required Lack of Essential Medicine
<u>Montreux</u>	Nutrition Food	04/07/2000 28/06/2000	Lack of Milk Distribution Required
<u>Roile</u>	Agricultural input	13/07/2000	Monitoring Required
<u>Zug</u>	Food Health Security Agricultural input Shelter	28/06/2000 03/07/2000 04/07/2000 27/06/2000 28/06/2000	Distribution Required Lack of Essential Medicine Camp 12 problems IDPs Monitoring Required Inadequate Structures

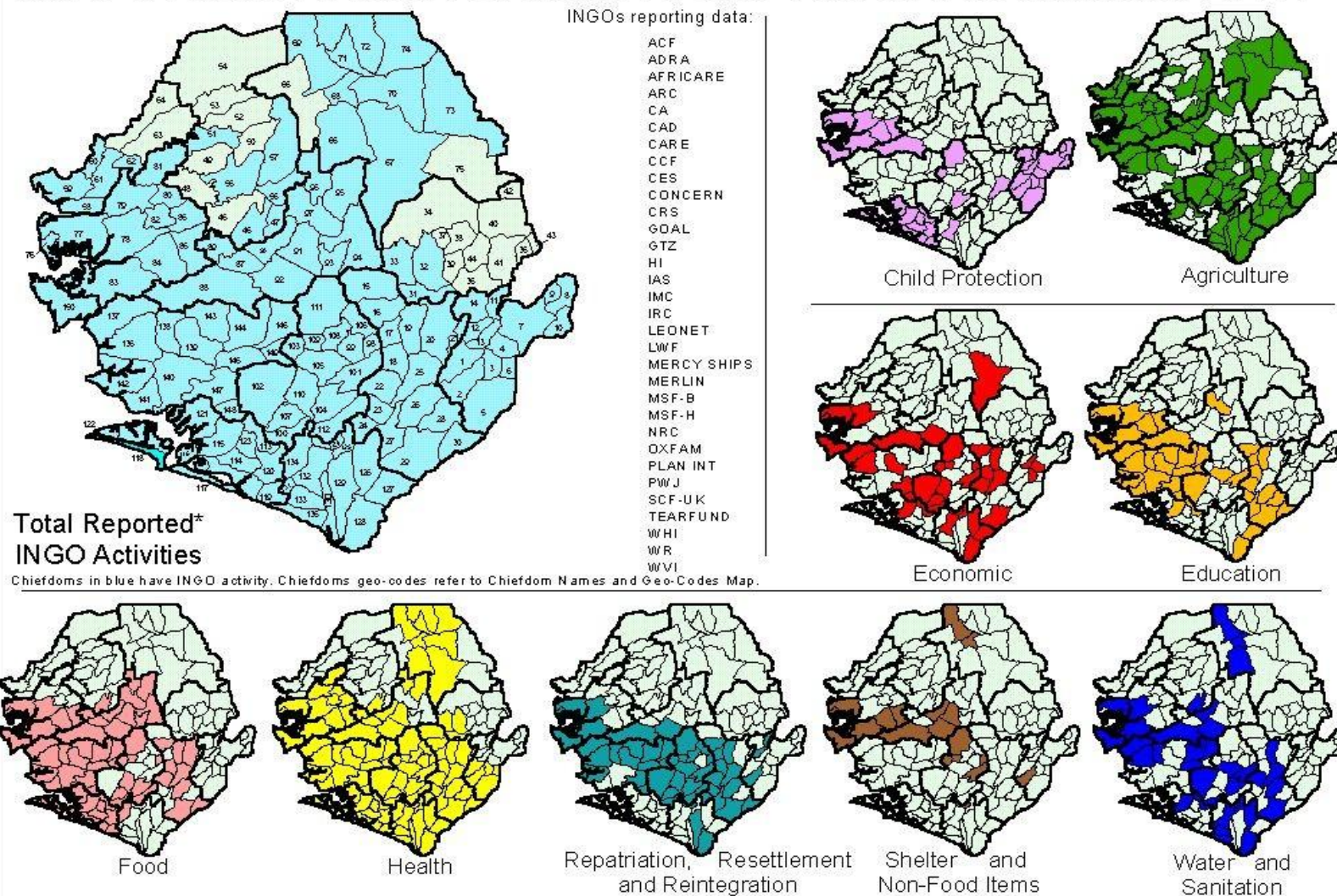
Wednesday, 26 October, 2000

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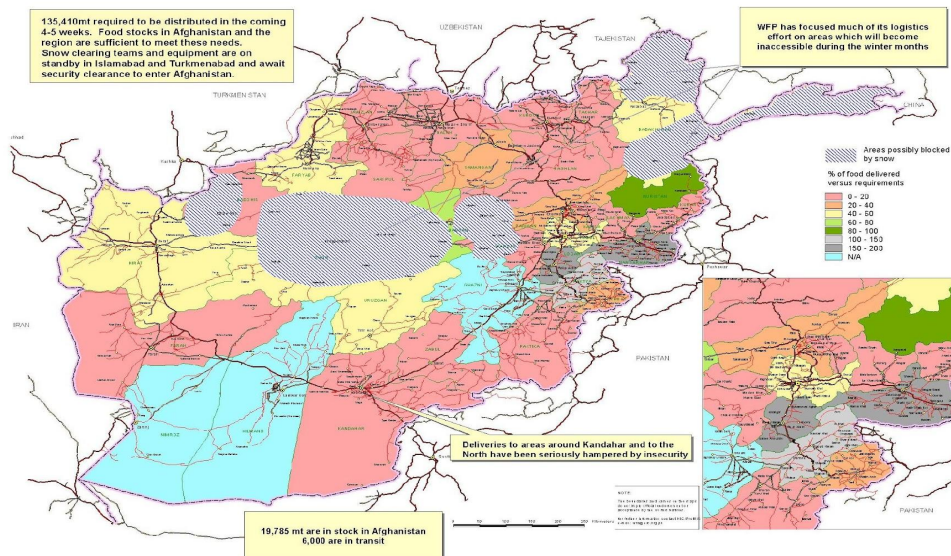
Sectoral Activities

INGO Sectoral Activities in Sierra Leone: Overview at Chiefdom Level

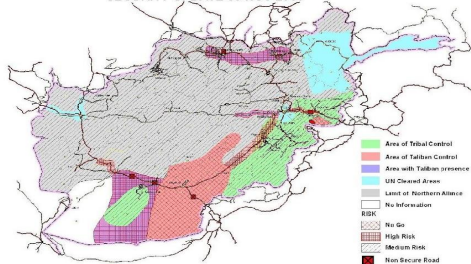


Food Deliveries

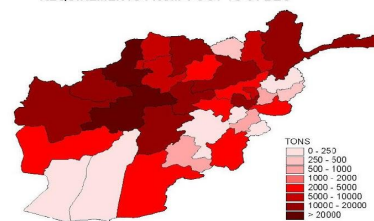
WFP Food Deliveries vs. Requirements: Oct-Dec 2001



SECURITY UPDATE 30 NOV 2001

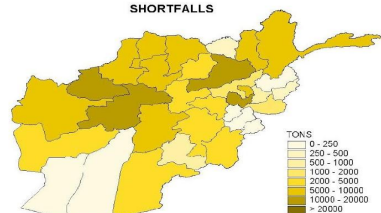


REQUIREMENTS FROM 1 OCT TO 31 DEC



Total to be delivered before year end: 135,410
In stock in Afghanistan or in transit: 25,785
In stock in the region: 114,569

SHORTFALLS



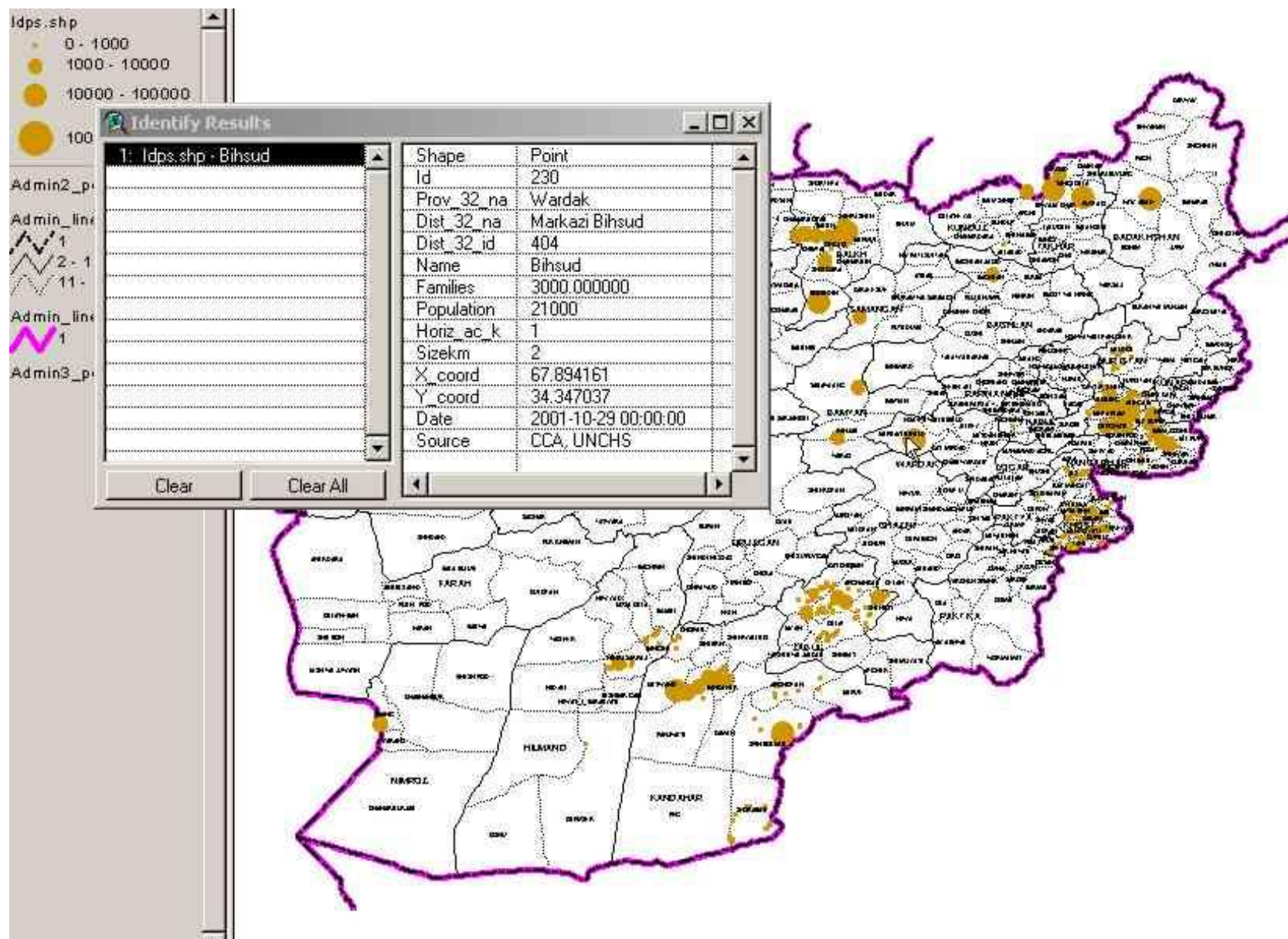
If WFP is to meet December delivery targets, the following is required:

- Security clearance for WFP trucks and drivers to operate in the country
- Deployment of snow clearing teams and equipment which are on standby awaiting security clearance.
- Establishment of three base camps in Bamyan, Panjao and Chaghchayan

DELIVERY AS PER 30 NOV



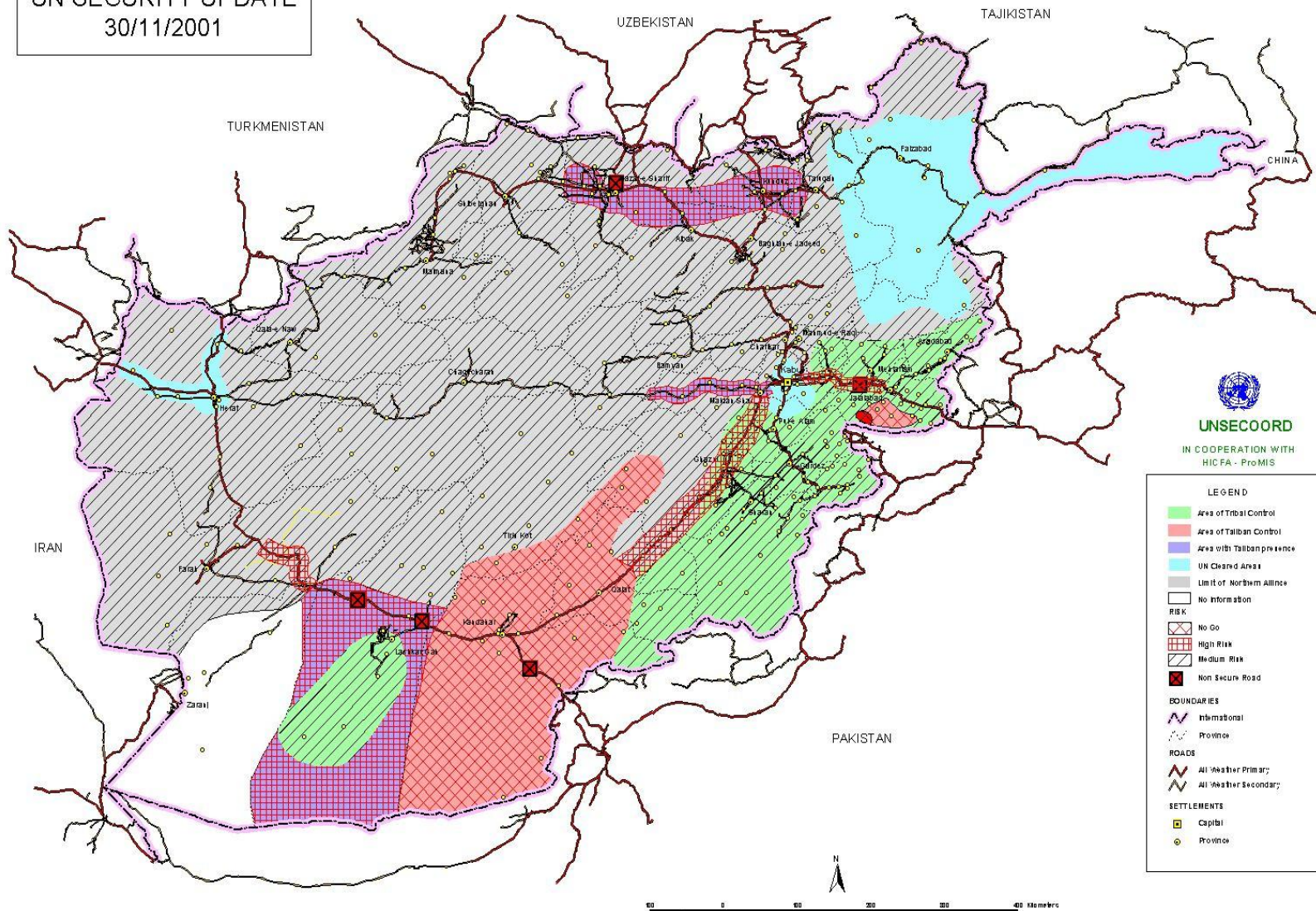
Internally Displaced Persons





UN Security Map

AFGHANISTAN
UN SECURITY UPDATE
30/11/2001



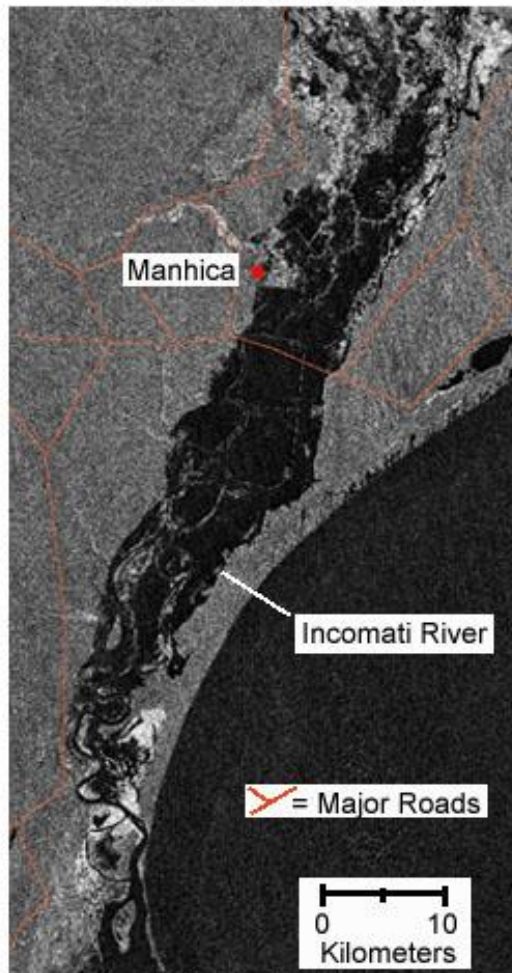
Possible outputs: Flooding Impact



Southern Mozambique Flooding Near Manhica (Radarsat/Landsat 7 Data Fusion)



RADARSAT 2/21/00



Source: Canadian Space Agency

LANDSAT 7/RADARSAT Merge



LANDSAT 7 7/5/99

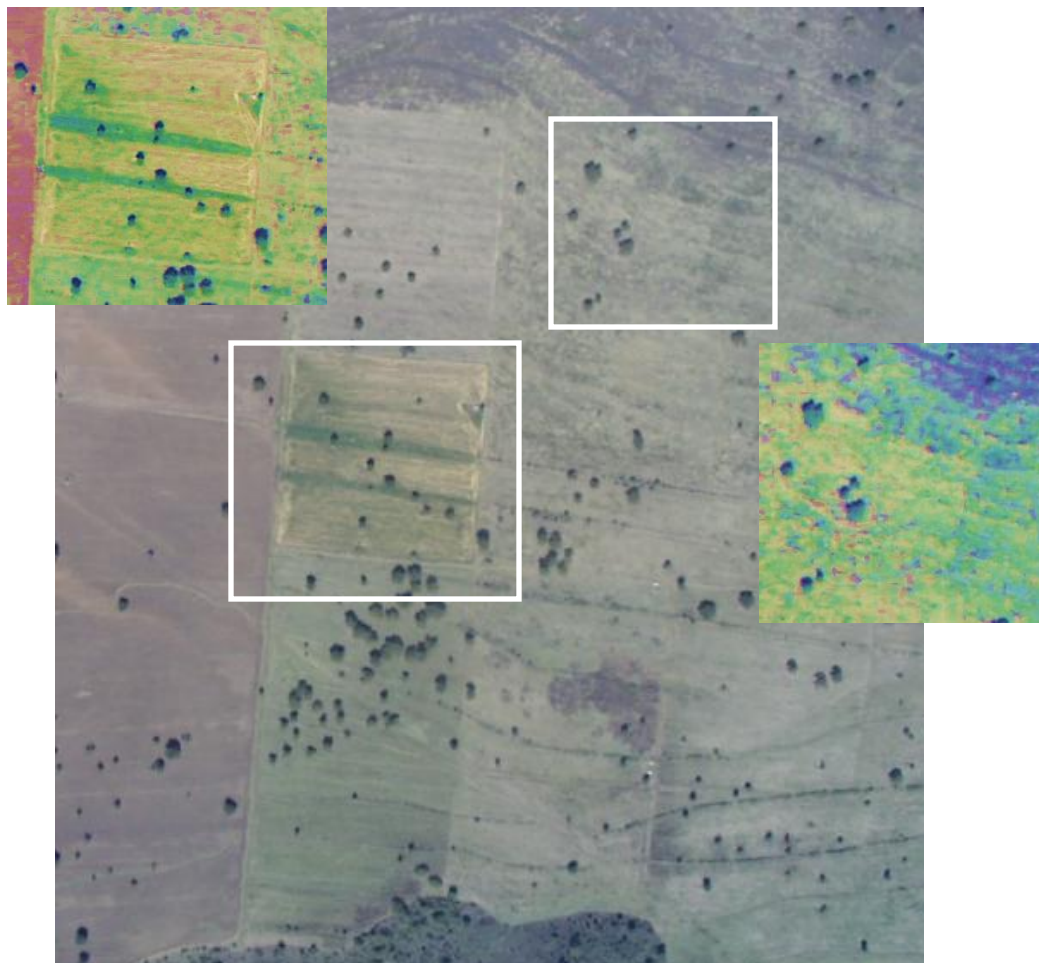


Source: EROS Data Center

Vegetation Monitoring

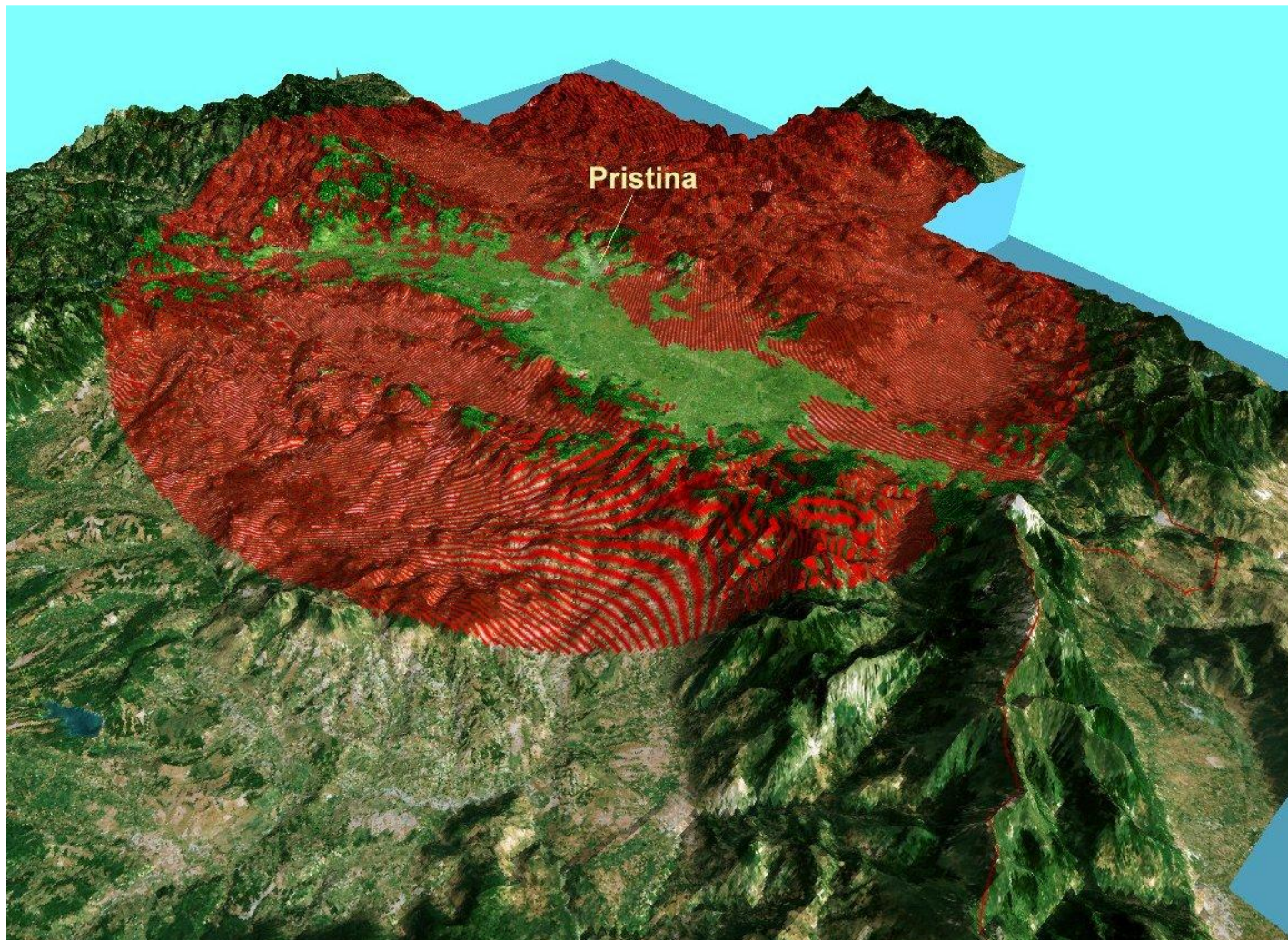
It is possible to highlight the presence of vegetation by appropriately filtering images on visible spectrum.

Low cost technology allows the acquisition of time series.



Radio communication coverage:

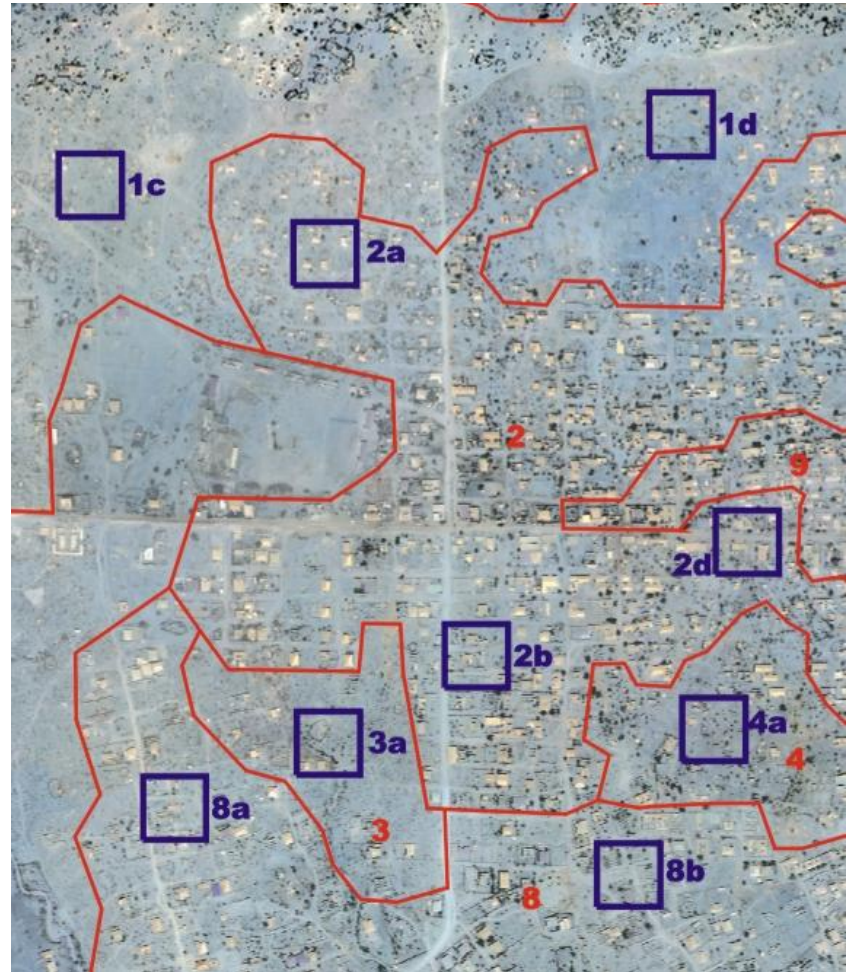
Fictive 15 m radio-mast placed in Pristina and corresponding coverage within reception area



Population Estimate

Population estimate has been done as follows:

- homogeneous house type zoning (red lines)
- dwelling units counting in sampling area (blue squares)
- extrapolation to the whole town



Accessibility to Services

Geo-referenced mosaics can be used by GIS packages to carry out spatial analysis.

Accessibility to water is just one of the many possible analysis.



Houses within a radius of 100 m from water sources

IFO Refugee Camp, Kenya

Individual trees planted inside blocks to provide shade

Individual shelters





Ifo Shelter Locations



Map

Database

Shelter	
ID	
Country code	
District name	
Location name	
Longitude DMS	
E/W	
Latitude DMS	
N/S	
Longitude DX	
Latitude DY	
Info Lat/Long	
Total Population (camp)	
Block name	

Country	
ID	
Continent	
Country name	
Country code	

Blocks	
ID	
Protection level	
Name	
Surface	
Type	
CountryCode	

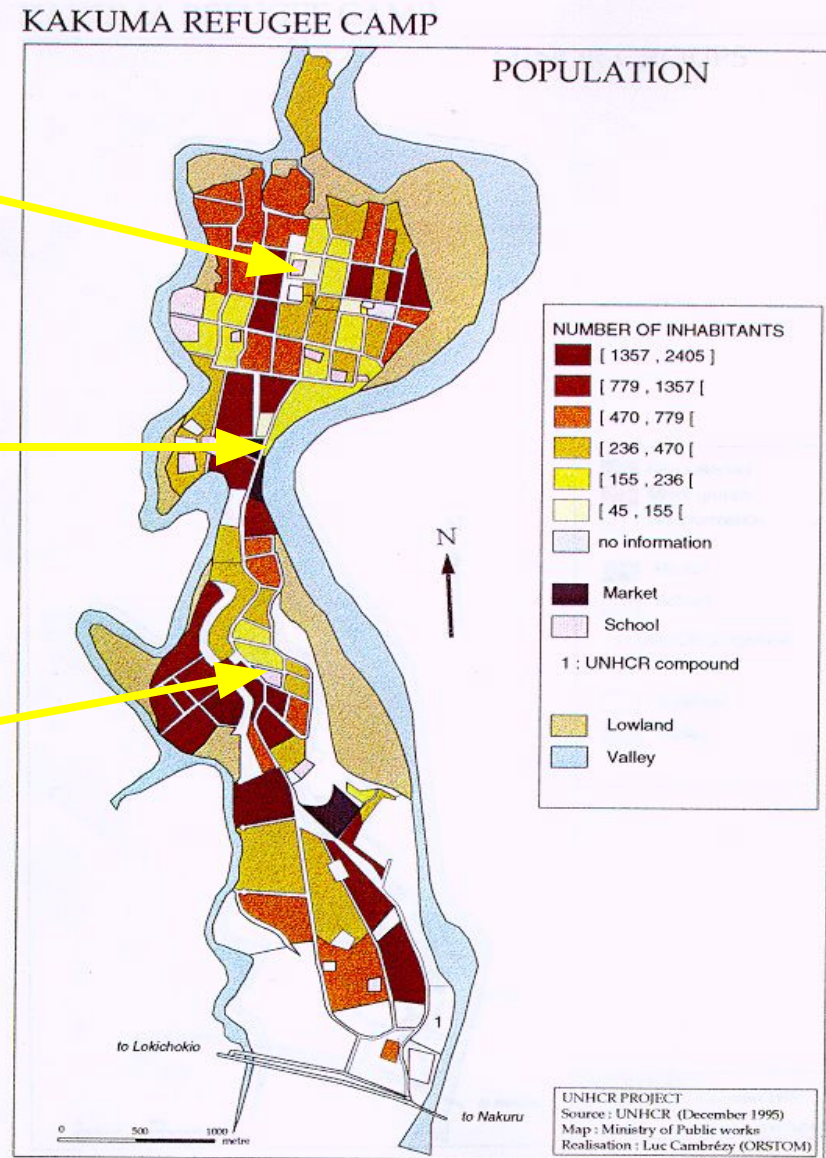
Offices	
ID	
COUNTRY CODE	
TYPE	
OFFICE NAME	
LONGITUDE DX	
LATITUDE DY	
INFO LAT/LONG	
REGIONAL COVERAGE	
COMMENTARY	

Kakuma Refugee Camp

**DENSITY
OF POPULATION**

**RIVER BANK
EROSION**

SCHOOLS





The End

GIST Web Site

<https://gist.itos.uga.edu/>

HIC Web Site

<http://www.humanitarianinfo.org>